Gravitational potential energy

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Learning target: Relate height change to near-surface gravitational energy.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 4.*
W_{net}=-12 J; speed falls for fixed mass.

02. Worked example: explain the model
Δh=3 m, ΔU=60 J; absolute heights may use any common reference.

03. Read and interpret the model
Higher book; for equal mass, energy proportional to height.

04. Apply the physics
U at 1 m = 20 J; U at 3 m = 60 J. Lifting from 1 m to 3 m gives ΔU = +40 J.

05. Explain your reasoning
No. It means location below chosen reference; differences matter.

06. Review: connect an earlier idea *Review: Unit 3, Day 9.*
They exchange velocities: A=-4 m/s; B=+6 m/s.

07. Exit check
60 J.

08. STRETCH: test the model
Examples 1 kg raised 6 m and 3 kg raised 2 m; positive change 60 J.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 4.*
$W_{net}=½m(v_f^2-v_i^2)=-36$ J, negative as kinetic energy falls.

02. Worked example: explain the model
$\Delta h=3$ m, $\Delta U=60$ J; absolute heights may use any common reference.

03. Read and interpret the model
Higher book; for equal mass, energy proportional to height.

04. Apply the physics
Floor reference: U_i = 200 J, U_f = 350 J, ΔU = +150 J. New reference: heights 2 m and 5 m, energies 100 J and 250 J. ΔU stays +150 J.

05. Explain your reasoning
No. It means location below chosen reference; differences matter.

06. Review: connect an earlier idea *Review: Unit 3, Day 9.*
$v_f=1$ m/s; $K_i=104$ J, $K_f=4$ J. K decreases by 100 J; perfectly inelastic.

07. Exit check
$\Delta U=-80$ J; gravity work +80 J.

08. STRETCH: test the model
$U_i=-60$ J, $U_f=150$ J; $\Delta U=210$ J; gravity does -210 J.